

TEAM VORTEX

DarkSky AI

Analyzing & Predicting Light Pollution in Baghdad

Identifying Candidate Low-Light Areas for Astronomical Observation

Team members :
Noor Ahmed Radhi
Alaa Qais Jasim
Alhussain Qassim Hussein
Hassan Alaa Nasser

Meet Team **Vortex**

A multidisciplinary team combining Geospatial Analysis, AI, and Strategy.

Alaa – ML & Geospatial Specialist

AI model training, spatial mapping & visual insights.

Alhussain – Video Producer & Editor

Video synthesis, visual storytelling & project presentation.

Noor – Geospatial Analyst

Spatial analysis, Python mapping & visual insights

Hassan – Project Strategist

Project synthesis, pitching & presentation design

The Problem

Baghdad suffers from widespread light pollution.

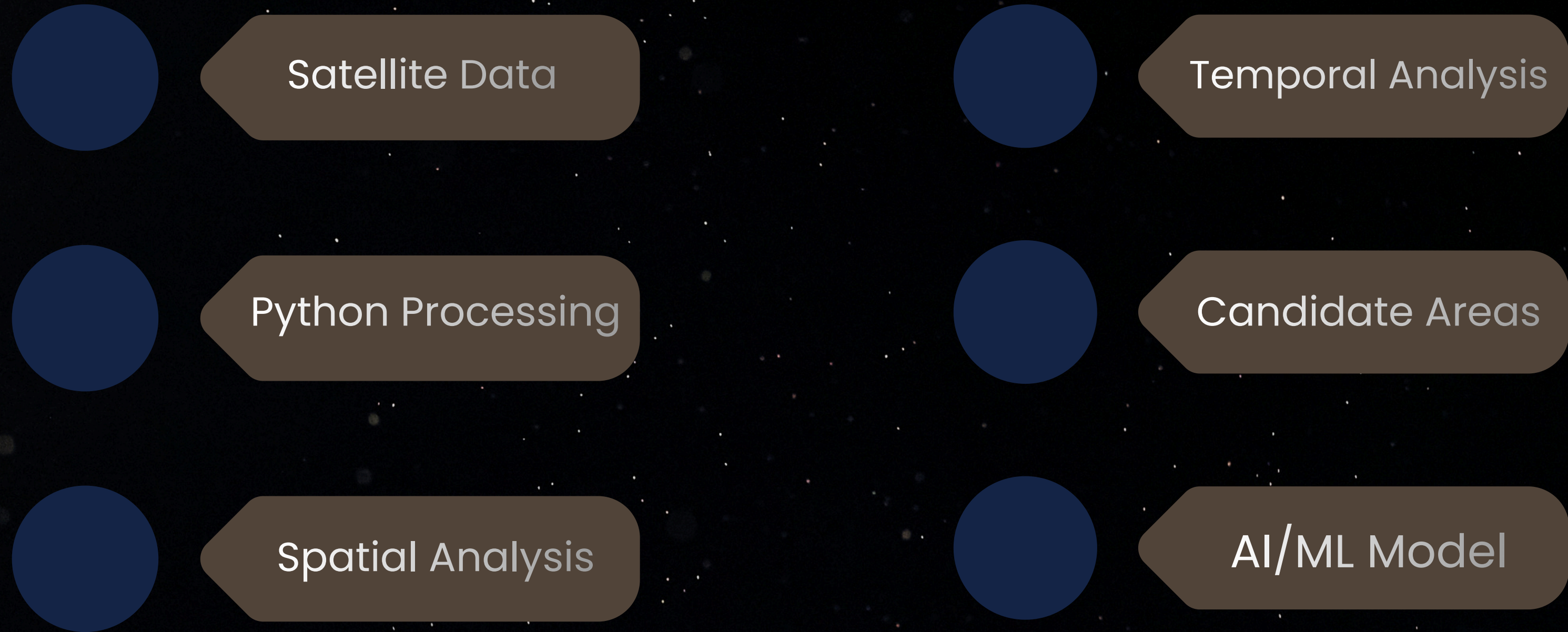
High radiance degrades night sky quality for astronomy.

Identifying dark zones is difficult using only visual impressions.

A data-driven approach is essential.



Our Solution



A comprehensive data pipeline to identify and predict optimal astronomical sites.

Dataset & Quality

19,600

VIIRS Observations

0

Missing or Duplicate Rows

36 Months

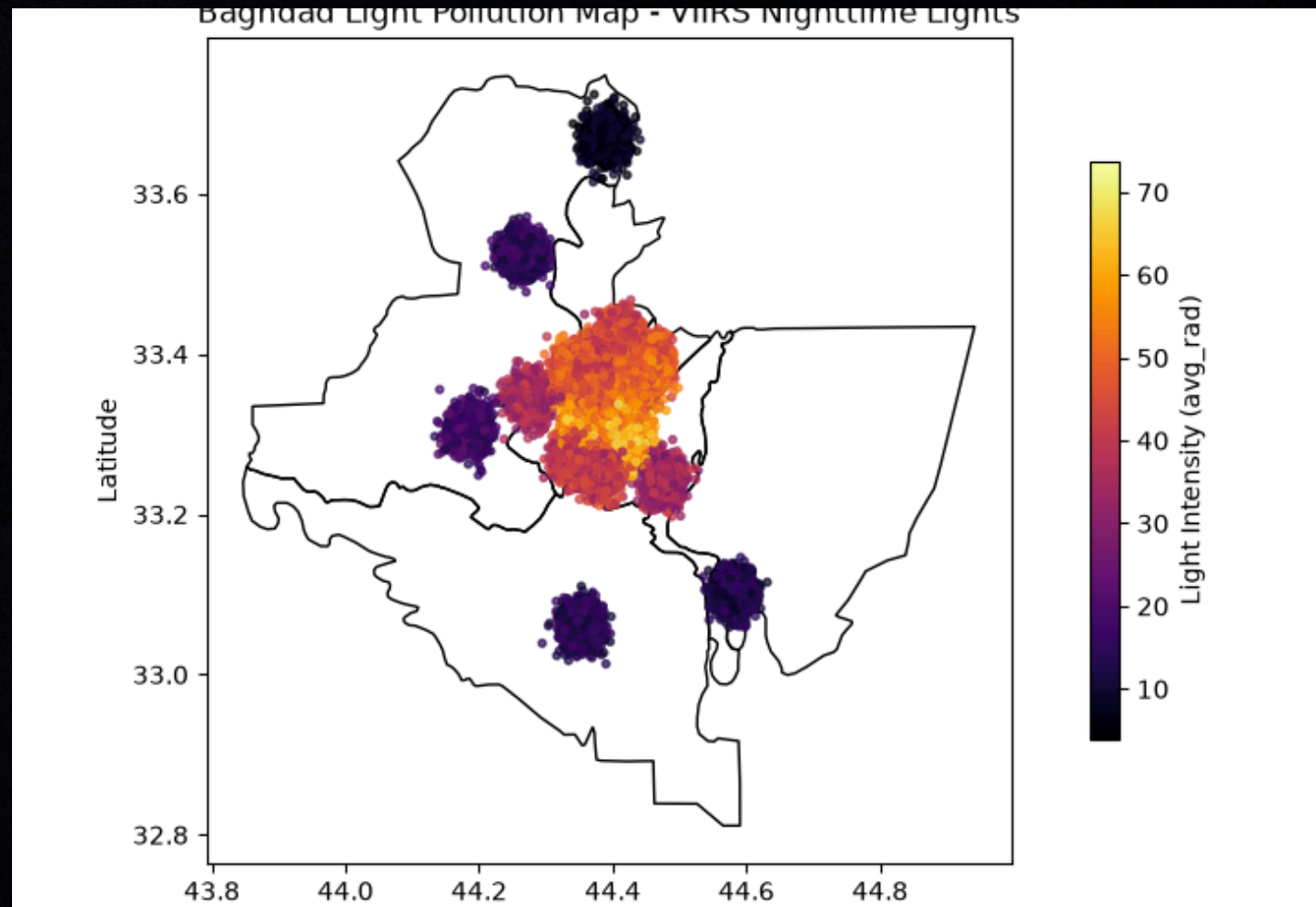
Coverage (2022–2024)

Baghdad

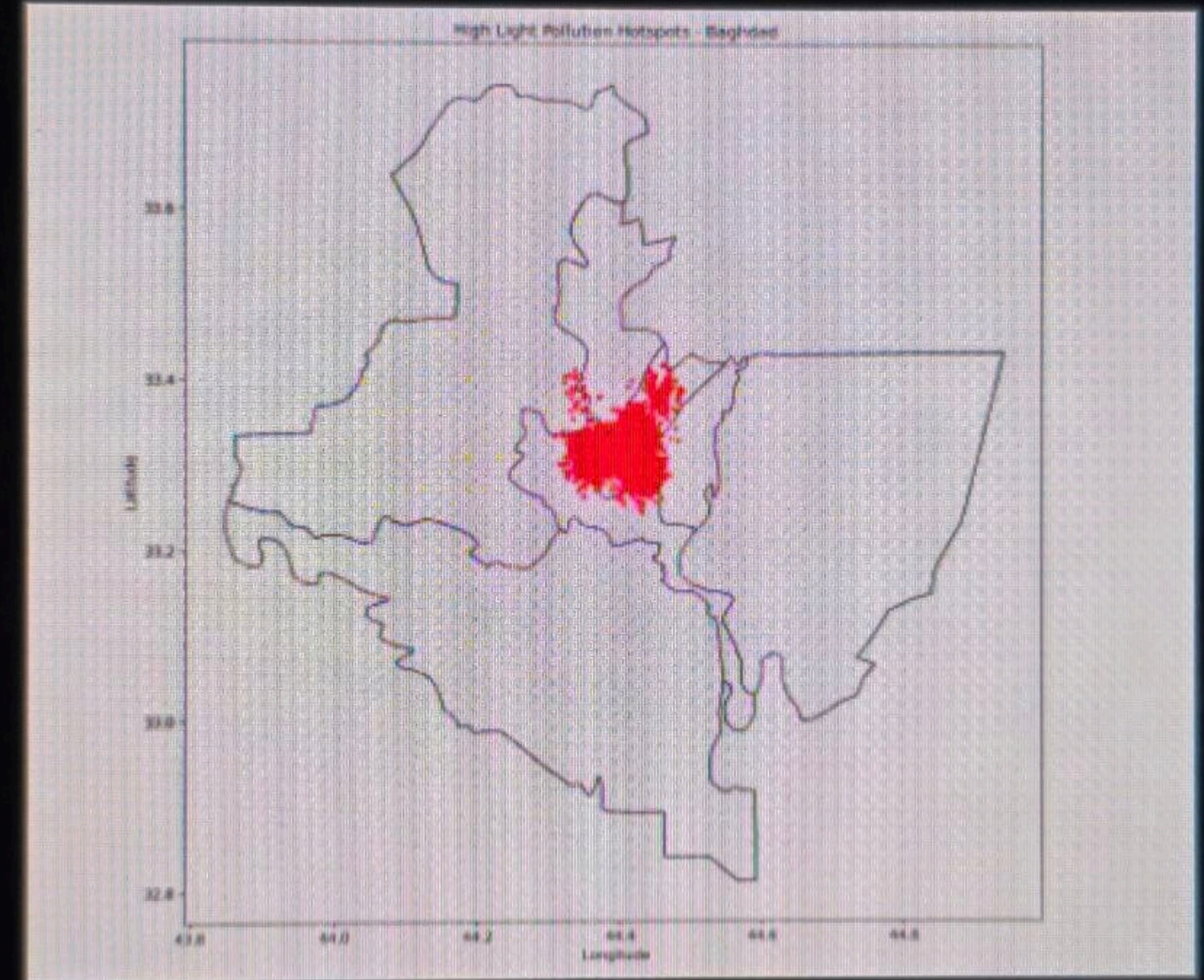
Geographic Coverage

Key Features: Latitude, Longitude, Average Radiance (avg_rad), Cloud-Free Coverage (cf_cvg).
Processed and validated via Excel & Python.

Spatial Analysis: Light Pollution Distribution



Hotspots: 1,963 Observations
(Identified using the 90th percentile)



Analyzed using VIIRS data. Higher values = Higher nighttime radiance.

Final Candidate Dark-Sky Areas

Filtered 869 grid cells down to 8
consistent low-light cells (4 Areas).

Group	2024 avg_rad	Months
Group 1	6.95	25
Group 2	7.29	14
Group 3	7.89	17
Group 4	10.95	23

Baghdad Stable-Grid Mean (2024): 34.83 avg_rad.
All candidate areas are substantially below the overall mean!

AI Validation Layer (Random Forest Regression)

Random Split

$R^2 = 0.92$

MAE: 3.70

Time-Based Split

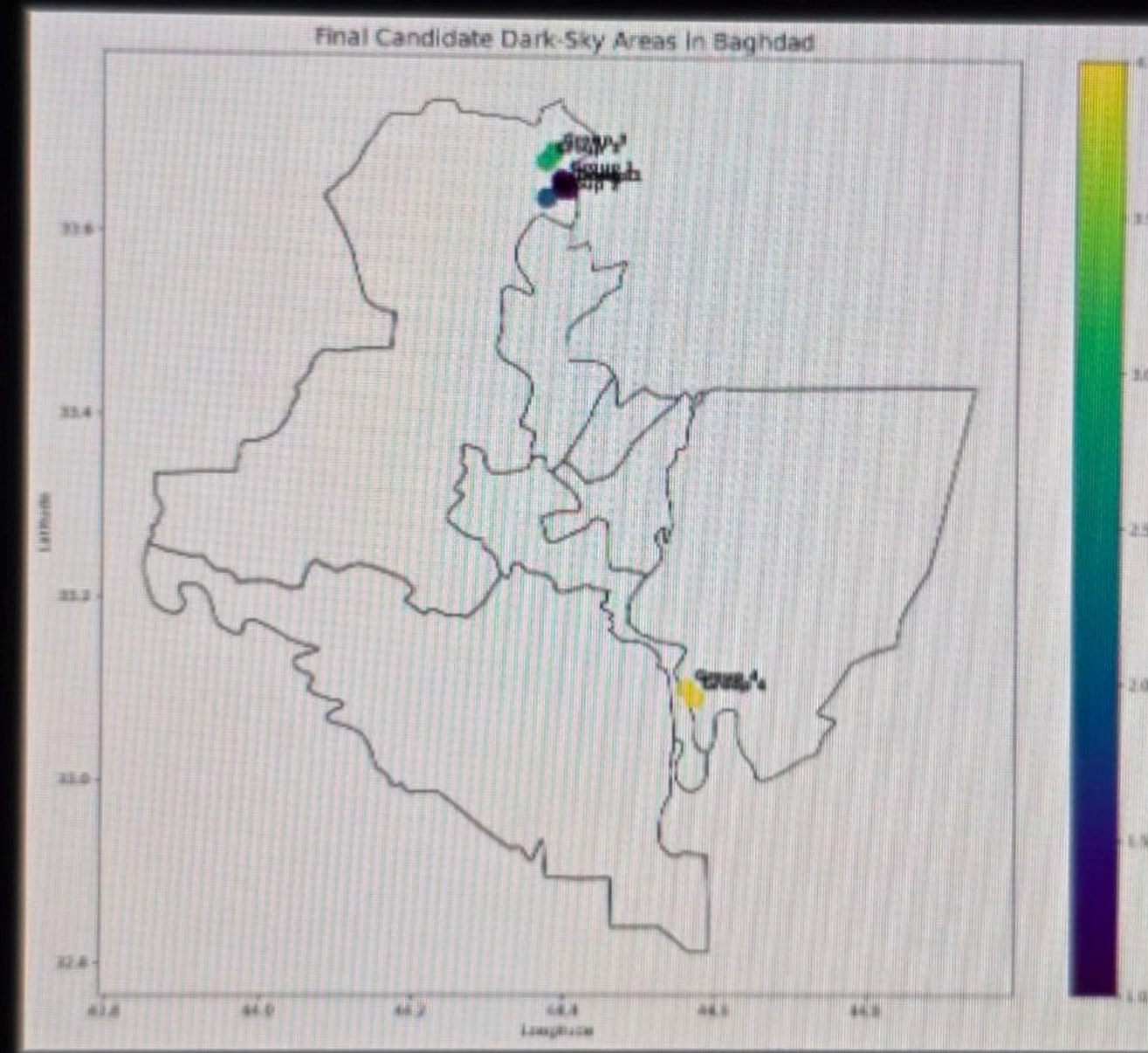
$R^2 = 0.90$

MAE: 4.12

Grid Cell	Actual 2024	Predicted ML
33.65, 44.39	6.5	7.8
33.64, 44.37	7.29	7.03
33.69, 44.38	8.06	8.2
33.10, 44.56	10.7	12.15

Model Features: Latitude, Longitude, Year, Month, cf_cvg.
Used as an AI-assisted consistency check.

Interactive Prototype & Heatmap



Folium & Leaflet.js | Esri Satellite Imagery
Spot #3 (Radiance: 6.50) – Validated via ML

Tech Stack & Libraries

- Python (Pandas, NumPy)
- Spatial: GeoPandas, GIS
- ML: Scikit-learn (Random Forest)
- Viz: Matplotlib, Folium, Leaflet

* AI (LLMs) was utilized strictly for workflow optimization and code debugging. All data outputs are natively computed.

Limitations & References

NASA/NOAA – VIIRS (Earth Observation Group)

HDX UN Boundaries

Esri World Imagery

- VIIRS measures satellite radiance, not exact ground sky quality.
 - Candidate areas require strict field validation.

Thank You

